

H General Utility Evaluation

We evaluate the general utility of steering methods on 200 privacy-irrelevant Alpaca instructions (Taori et al., 2023), scoring steered responses with GPT-4o-mini for helpfulness, coherence, and relevance (1–5 scale).

Table 5: Utility evaluation on 200 Alpaca instructions (Llama-3.1-8B).

	α	Help.	Coh.	Rel.
No Steering	0	4.58	4.76	4.89
Monolithic	0.5	4.13	4.61	4.62
CI-Parametric	0.5	1.04	2.53	1.51
Monolithic	1.0	1.07	2.40	1.28
CI-Parametric	1.0	1.00	1.00	1.00

I Five-Parameter CI Ablation

Contextual Integrity defines five parameters: data subject, sender, recipient, information type, and transmission principle. Our main experiments use only the three *norm-determining* parameters (information type, recipient, transmission principle). We validate this design choice by extending CI-parametric steering to all five parameters and comparing against the three-parameter variant.

Setup. We generate CI-decomposition probing examples for all five parameters (2,500 total; 500 per parameter), extract per-parameter directions, and evaluate the full pipeline on synthetic data (200 balanced scenarios), CONFAIDE Tier 3, and PrivaCI-Bench. All other hyperparameters remain unchanged.

Results on Synthetic Data. Table 6 compares three- and five-parameter steering on Llama-3.1-8B at $\alpha=1.0$. The three-parameter variant nearly eliminates leakage (0.5%, PPI = 98.8%), whereas the five-parameter variant achieves only moderate reduction (16.0%, PPI = 62.4%).

The ablation pattern is more revealing. With three parameters, each individual axis independently reduces leakage (PPI between 35.3% and 56.5%). With five parameters, *every* individual axis increases leakage beyond the unsteered baseline (all PPIs negative), indicating that the additional directions introduce noise that disrupts existing privacy behavior.

Table 6: Three- vs. five-parameter CI steering on synthetic data (Llama-3.1-8B, $\alpha=1.0$).

Steering Axis	3-Param		5-Param	
	Leak. $\downarrow$	PPI $\uparrow$	Leak. $\downarrow$	PPI $\uparrow$
No Steering	.425	—	.425	—
Info Type only	.230	.459	.500	-.176
Recipient only	.185	.565	.530	-.247
Trans. Principle only	.275	.353	.495	-.165
Data Subject only	—	—	.490	-.153
Sender only	—	—	.465	-.094
All CI params	.005	.988	.160	.624

Results on CONFAIDE and PrivaCI-Bench. Table 7 reports transfer results. On CONFAIDE, the five-parameter variant achieves 0.0% leakage at $\alpha=0.5$ (vs. 2.6% for three parameters) but degrades to 16.7% at $\alpha=1.0$ (vs. 0.0% for three parameters), losing the desirable monotonic reduction. On PrivaCI-Bench, five-parameter steering reaches 8.0% leakage at both α values, while the three-parameter variant eliminates leakage entirely at $\alpha=1.0$. The

inability of the five-parameter variant to reach zero leakage at higher α suggests interference rather than reinforcement from the additional directions.

Table 7: Results on CONFAID and PrivaCI-Bench (Llama-3.1-8B). Five-parameter steering shows marginal gains at low α but fails to eliminate leakage at higher α .

Dataset	α	3-Param Leak.↓	5-Param Leak.↓
CONFAIDE	0.5	.026	.000
	1.0	.000	.167
PrivaCI	0.5	.153	.080
	1.0	.000	.080

Analysis. The strongest failure occurs on Llama-2-7B: five-parameter steering at $\alpha=1.0$ increases synthetic leakage from 52.0% to 88.0% (PPI = -69.2%). On CONFAIDE, leakage rises to 70.4% (vs. 43.3% for three parameters at the same α). This fragility indicates that data-subject and sender directions, while geometrically identifiable, do not encode stable norm-relevant signals across architectures.